

Applied Optics for Fiber Optic Communications: Theory and Practice – Part 3

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Date : April 2025

Contents

Chapter 3: Wavelengths in Fibre optical communications	3
3.1 Introduction to Wavelengths	3
3.1.1 Wave-Particle Duality	3
3.1.2 Examples.....	4
3.2 How Wavelength Relates to Optical Fiber Communication.....	5
3.3 Wavelengths and the Electromagnetic Spectrum	6
3.4 Chromatic Dispersion	9
Chapter 4: Basics of Fibre Optical Communications	10
4.1 Fibre Optical Communication	10
4.2 Key Components	10
4.3 Key Technologies	10
Chapter 5 : DWDM.....	11
5.1 Key Features of DWDM.....	11
5.2 How DWDM Works.	13
5.3 What are the DWDM Components?	13
5.4 How Does DWDM System Components Work Together with DWDM Technology?	26
5.5 DWDM Topologies	27
5.6 DWDM Network Security / Protections	30

Chapter 3: Wavelengths in Fibre optical communications

3.1 Introduction to Wavelengths

A wavelength is the distance between successive peaks (or troughs) of a wave. It is a fundamental property of waves, including light waves used in optical fiber communication. Wavelength is typically measured in nanometers (nm) for optical signals, where 1 nm equals 10^{-9} meters.

In simpler terms, imagine waves in the ocean—if you measure the distance from one crest to the next crest, that's the wavelength. Similarly, for light traveling through an optical fibre, the wavelength refers to the distance between peaks of the electromagnetic wave.

Wavelength is inversely related to frequency, which is the number of wave cycles that pass a point per second. This relationship is governed by the equation:

$$\lambda = c / f$$

Where:

- λ : Wavelength (in meters).
- c : Speed of light (3×10^8 m/s).
- f : Frequency (in Hz).

For example, light with a frequency of 200 THz (terahertz) has a wavelength of approximately 1500 nm, which falls within the C-band commonly used in DWDM systems.

3.1.1 Wave-Particle Duality

Light exhibits both wave-like and particle-like behaviour, known as *wave-particle duality*. While wavelength describes its wave nature, energy relates to its particle nature. The energy (E) of a photon is directly proportional to its frequency (f) and inversely proportional to its wavelength (λ):

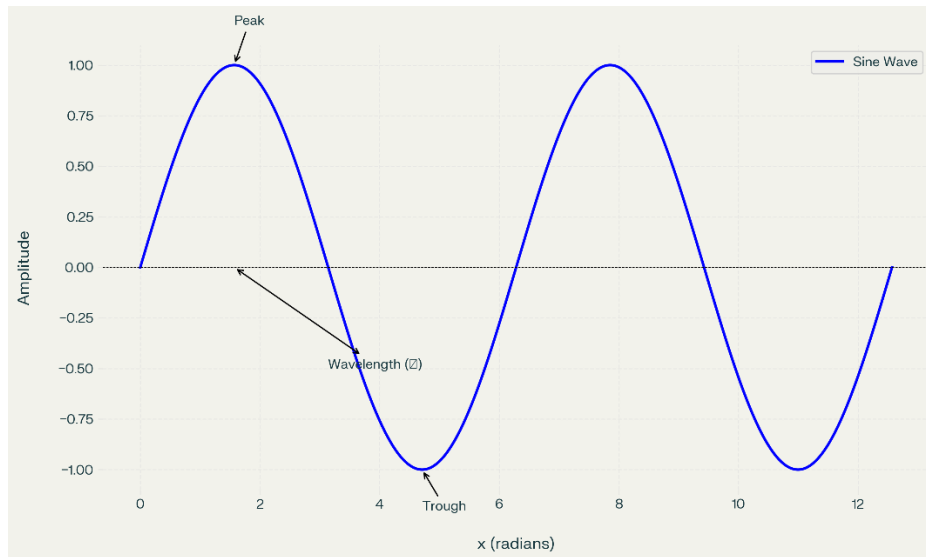
$$E = h \cdot f = \frac{h \cdot c}{\lambda}$$

Where:

- E : Energy (in joules).

- h : Planck's constant (6.626×10^{-34} Js).

This relationship explains why shorter wavelengths (e.g., ultraviolet light) have higher energy than longer wavelengths (e.g., infrared light).



The above image illustrates A sine wave showing the peak, trough, and wavelength (λ). The wavelength is the distance between successive peaks or troughs

3.1.2 Examples

For calculating wavelengths in optical fibre communication using the equation/ formula $\lambda = c / f$, where $c = 3 \times 10^8$ m/s (speed of light), λ is wavelength, and f is frequency:

Scenario 1 : A DWDM system uses a frequency of **193.1 THz** . Calculate the wavelength.

Formula :

$$f = \frac{c}{\lambda} = \frac{3 \times 10^8 \text{ m/s}}{1310 \times 10^{-9} \text{ m}}$$

Solution:

$$f = \frac{3 \times 10^8}{1.31 \times 10^{-6}} = 2.29 \times 10^{14} \text{ Hz} = 229 \text{ THz}$$

Scenario 2 : A fibre link uses **1310 nm** . Calculate the frequency.

Formula:

$$f = \frac{c}{\lambda} = \frac{3 \times 10^8 \text{ m/s}}{1310 \times 10^{-9} \text{ m}}$$

Solution :

$$f = \frac{3 \times 10^8}{1.31 \times 10^{-6}} = 2.29 \times 10^{14} \text{ Hz} = 229 \text{ THz}$$

3.2 How Wavelength Relates to Optical Fiber Communication

In optical fibre communication:

- **Shorter wavelengths** tend to experience higher attenuation and dispersion but can carry more energy.
- **Longer wavelengths**, are ideal for long-distance communication due to their low attenuation and compatibility with EDFAs (Erbium-Doped Fiber Amplifiers).

Fiber optic communication relies on specific wavelengths of light to transmit data efficiently over long distances. These wavelengths are chosen based on their transmission properties, such as low attenuation, minimal dispersion, and compatibility with amplifiers

3.3 Wavelengths and the Electromagnetic Spectrum

3.3.1 Electromagnetic Spectrum Overview

The electromagnetic spectrum is the complete range of electromagnetic radiation, organized by frequency, wavelength, or photon energy. Electromagnetic radiation consists of waves of electric and magnetic fields traveling through space at the speed of light ($3 \times 10^8 \text{ m/s}$) in a vacuum. The electromagnetic spectrum comprises seven primary bands, each with unique properties and applications. The Seven Bands of the Electromagnetic Spectrum:

1. Gamma Rays

- Discovery: Henri Becquerel (1896).
- Production: Emitted during nuclear reactions (e.g., radioactive decay), not from electron transitions.
- **Relation to Fiber Optics:** Not used in optical fibre communication due to extremely high energy and ionizing nature, which damages materials.

2. X-Rays

- Discovery: Wilhelm Röntgen (1895).
- Production: Generated by bombarding high-atomic-number materials (e.g., tungsten) with high-energy electrons.
- Key Property: Penetrate soft tissue but not bones.
- **Relation to Fiber Optics:** Not used in fibre optics; their high energy disrupts fibre structure

3. Ultraviolet (UV) Rays

- Discovery: Johann Ritter (1801).
- Production: Emitted by excited atoms/molecules or celestial bodies like the Sun.
- **Relation to Fiber Optics:** Rarely used due to high attenuation in silica fibres. UV curing is sometimes used in fibre coating processes.

4. Visible Light

- Characteristics: The only band visible to humans, spanning 400–700 nm (violet to red).
- Production: Generated by excitation of valence electrons (e.g., LEDs, lasers).
- **Relation to Fiber Optics:**
 - Multimode fibres sometimes use visible light (e.g., 650 nm for short-distance communication).
 - Plastic optical fibres (POF) utilize red light (650 nm) for low-cost applications like automotive networks.

5. Infrared (IR) Waves

- Characteristics: Wavelengths between 700 nm and 1 mm, often called "heat radiation."
- Production: Emitted by hot bodies or molecular vibrations.
- **Relation to Fiber Optics:**
 - Primary band for fibre communication:
 - **850 nm:** Used in multimode fibres for short-range links (data centres).
 - **1310 nm and 1550 nm:** Standard wavelengths in single-mode fibres for long-haul communication (minimal attenuation, EDFA compatibility).
 - Governs modern DWDM and OTN systems.

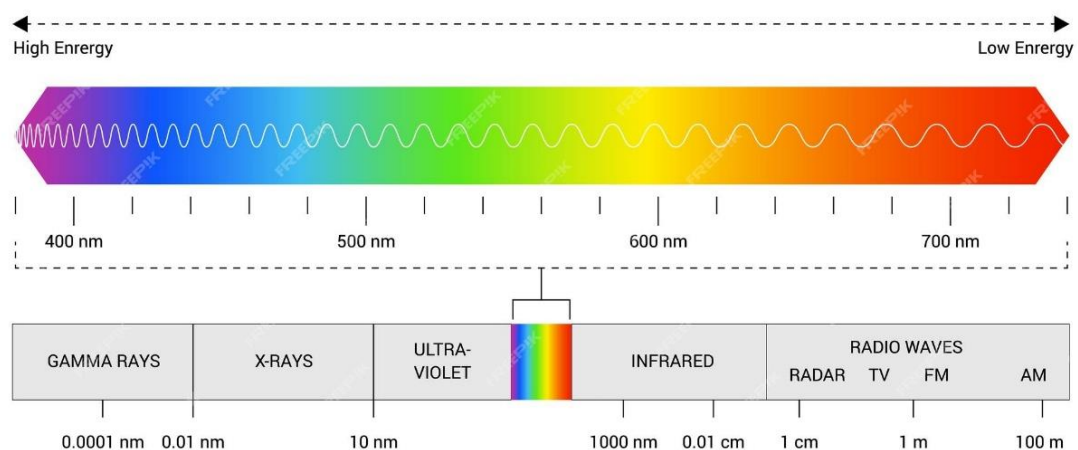
6. Microwaves

- Production: Generated by oscillating electrons in circuits (e.g., magnetrons, klystrons).
- Applications: Radar, satellite communication, and microwave ovens.
- **Relation to Fiber Optics:** Not used directly, but fibre backhuls often carry microwave-generated data (e.g., 5G networks).

7. Radio Waves

- Characteristics: Longest wavelengths (1 mm to 100 km).
- Production: Produced by oscillating charges in antennas.
- **Relation to Fiber Optics:** Complementary to fibre: Radio waves (e.g., Wi-Fi, cellular signals) are converted to optical signals for long-distance transmission via fibre

VISIBLE SPECTRUM



Band	Relevance to Fiber Optics
Gamma Rays / X-Rays	Not used (damaging to materials).
UV Rays	Rarely used (high attenuation).
Visible Light	Limited use (POF, short-range multimode fibres).
Infrared	Core band(850 nm, 1310 nm, 1550 nm).
Microwaves / Radio	Indirect use (data conversion to optical signals).

- **Low Attenuation:** Minimal signal loss at 1310 nm and 1550 nm.
- **EDFA Compatibility:** Erbium-doped amplifiers boost signals in the C-band (1530–1565 nm).
- **DWDM Efficiency:** Allows dense packing of wavelengths (e.g., 40+ channels at 100 GHz spacing).

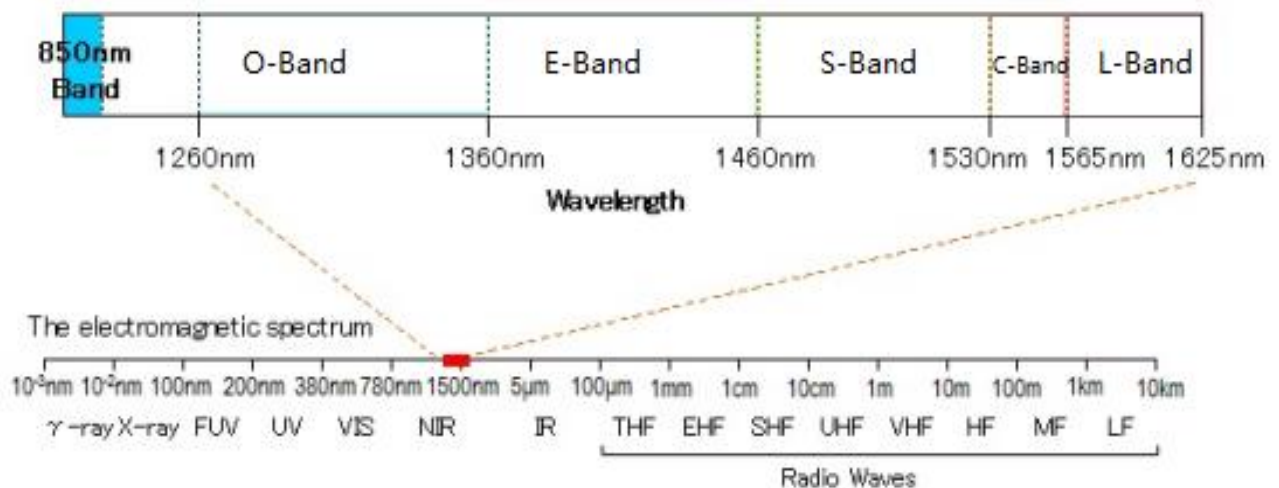
Example : A single-mode fibre link operates at a wavelength of 1550 nm over a distance of 100 km with an attenuation rate of 0.2 dB/km. What is the total attenuation?

Solution: **Total Attenuation** = **Attenuation Rate** × **Distance** = 0.2 dB/km × 100 km = **20 dB**

3.3.2 Optical Communication Bands:

To standardize wavelength usage in optical networks, the International Telecommunication Union (ITU-T) has defined specific bands:

Band	Wavelength Range (nm)	Applications
O-Band (Original band)	1260 - 1360	Early optical communication; minimal chromatic dispersion.
E-Band (Extended band)	1360 - 1460	Extended band; used with Zero Water Peak fibres for improved performance.
S-Band (Short band)	1460 - 1530	Short wavelength band; emerging applications in metro networks.
C-Band (Conventional band)	1530 - 1565	Widely used in DWDM systems; lowest attenuation and EDFA compatibility.
L-Band (Long band)	1565 - 1625	Extends DWDM capacity beyond C-band; used in long-haul and submarine systems.
U-Band (Ultra Long band)	>1625	Reserved for testing and maintenance without disrupting active traffic.



3.3.3 Practical Applications of Common Wavelengths

- **Multimode Fibers at 850 nm:**
 - Ideal for short-distance applications like connecting servers within data centres.
 - Example: Ethernet links operating at speeds up to 10 Gbps over multimode fibre.

- **Single-Mode Fibers at 1310 nm:**
 - Used in metro networks where moderate distances (<50 km) are required.
 - Example: GPON systems using upstream wavelengths at 1310 nm.
- **Long-Haul Systems at 1550 nm:**
 - Enables high-capacity DWDM systems for global communication.
 - Example: Submarine cables transmitting terabits of data across continents, Long haul networks.

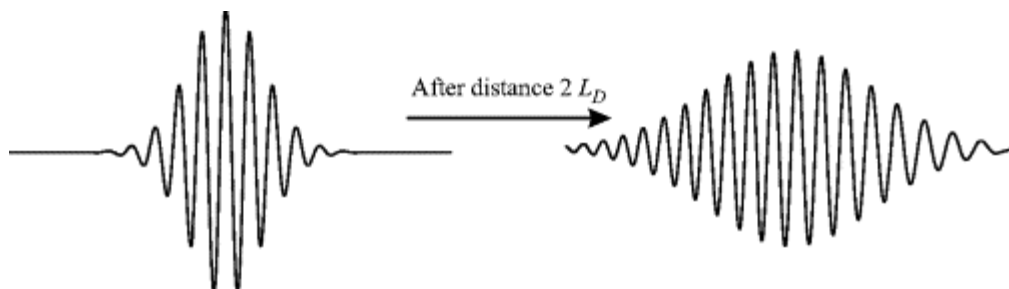
3.4 Chromatic Dispersion

Dispersion in optical fibres is a critical factor affecting signal quality and data transmission. Chromatic dispersion arises because different wavelengths of light travel at varying speeds within a medium like optical fibre.

This phenomenon has two components:

- **Material dispersion:** Caused by the wavelength-dependent refractive index of the fibre material (e.g., silica glass)
- **Waveguide dispersion:** Results from the fiber's core-cladding structure guiding light differently for various wavelengths

The combined effect causes pulse broadening, where a sharp input pulse gradually flattens



Key Components Shown in Figure:

- **Input Pulse:** Narrow Gaussian profile (20-50 ps width) representing digital "1" in optical communication
- **Broadened Pulse:** Flattened temporal profile after fiber propagation, demonstrating:
 - Increased full width at half maximum (FWHM)
 - Reduced peak power (same total energy)

Chapter 4: Basics of Fibre Optical Communications

4.1 Fibre Optical Communication

Fiber optic communication is a method of transmitting data as light signals through thin glass or plastic fibres. This technology has revolutionized telecommunications by offering high-speed data transmission over long distances with minimal signal degradation.

4.2 Key Components

Based on the concepts from Chapter 1 and Chapter 2, here are the components in fibre-optical communications.

- **Optical Fiber (Cable)**
 - **Core:** Guides light signals.
 - **Cladding:** Ensures light stays within the core.
 - **Coating (Buffer):** Protects the fibre from environmental damage.
- **Optical Transmitter**
 - Converts electrical signals into optical signals using light sources like laser diodes or LEDs.
- **Optical Receiver**
 - Converts incoming optical signals back into electrical signals.
- **Optical Amplifiers**
 - Boost the strength of optical signals to extend transmission distances.
- **Connectors and Splices**
 - Connectors join two fibres together, while splices permanently fuse them.

4.3 Key Technologies

- **DWDM (Dense Wavelength Division Multiplexing):** Allows multiple wavelengths to be transmitted over a single fibre, increasing network capacity.
- **OTN (Optical Transport Network):** Provides a framework for efficient data transport, enabling sub-lambda grooming and reducing bandwidth waste.

Chapter 5 : DWDM

Dense Wavelength Division Multiplexing (**DWDM**) is an advanced optical fibre multiplexing technology designed to significantly increase the bandwidth and data-carrying capacity of fibre optic networks. It works by combining multiple data signals, each transmitted on its own unique light wavelength, onto a single pair of optical fibres. This allows for the simultaneous transmission of numerous data streams such as voice, video, and data

- Without interference, all while maintaining complete separation between channels
- Each wavelength represents an independent optical channel

5.1 Key Features of DWDM

5.1.1 High Channel Density / Channel Spacing

Channel spacing in Dense Wavelength Division Multiplexing (DWDM) refers to the frequency or wavelength interval between adjacent optical channels transmitted over a single fibre. This spacing determines how many separate data channels can be multiplexed within the available optical spectrum.

Typical DWDM Channel Spacing Standards

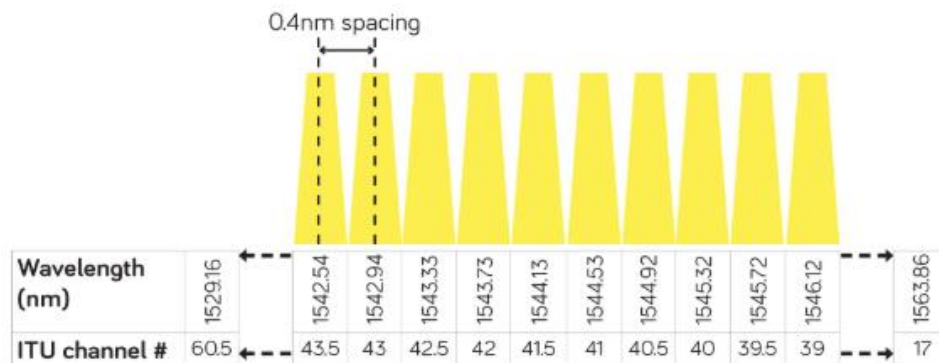
- **Common Standards:** The most widely used DWDM channel spacings are **100 GHz** and **50 GHz**, which correspond to approximately **0.8 nm** and **0.4 nm** wavelength separations, respectively, in the C-band (around 1550 nm)
- **Higher Density:** Some advanced systems and standards allow for even narrower spacing, such as **25 GHz** or less, especially with the introduction of flexible grid (Flex-Grid) technology, which enables dynamic adjustment of channel widths based on network requirements

Why Tight Channel Spacing?

- **Increased Capacity:** Tighter channel spacing means more channels can fit within the same fibre bandwidth, significantly increasing the total data-carrying capacity
- **Efficient Spectrum Use:** DWDM systems, by using narrow channel spacing, can accommodate up to 40, 80, 96, or even more channels in the C-band, each carrying high-speed data independently

For Example :

- A typical DWDM system of 0.4nm (50 GHz) spacing can create up to 80-160+ channels.



- A typical DWDM system of 0.8nm (100 GHz) spacing can create up to 40-80 channels.

Key Points:

- Standardized grids (e.g., ITU-T G.694.1) ensure interoperability between equipment from different vendors
- Amplification compatibility: DWDM channels are spaced to allow simultaneous amplification by optical amplifiers (such as EDFAs) without channel interference

5.1.2 Massive Bandwidth

By allowing each channel to carry high-speed data (e.g., 100 Gbps per channel), DWDM enables total fibre capacities in the terabits per second range

5.1.3 Long-Distance Transmission

DWDM signals can be amplified using optical amplifiers (such as Erbium-Doped Fiber Amplifiers, EDFAs), enabling data to travel over hundreds or thousands of kilometres without significant loss or the need for electrical regeneration

5.1.4 Efficient Fiber Utilization

DWDM maximizes the use of existing fibre infrastructure, reducing the need for laying new fibre cables and thus lowering costs and complexity for network operators

5.2 How DWDM Works.

5.2.1 Multiplexing

At the transmission end, a multiplexer combines multiple input data streams, each converted into a specific wavelength of light, onto a single optical fibre.

5.2.2 Transmission

The combined multi-wavelength signal travels through the fibre. Optical amplifiers boost the signal as needed for long-haul transmission

5.2.3 Demultiplexing

At the receiving end, a demultiplexer separates the combined signal back into individual wavelengths, each carrying its original data stream

5.3 What are the DWDM Components?

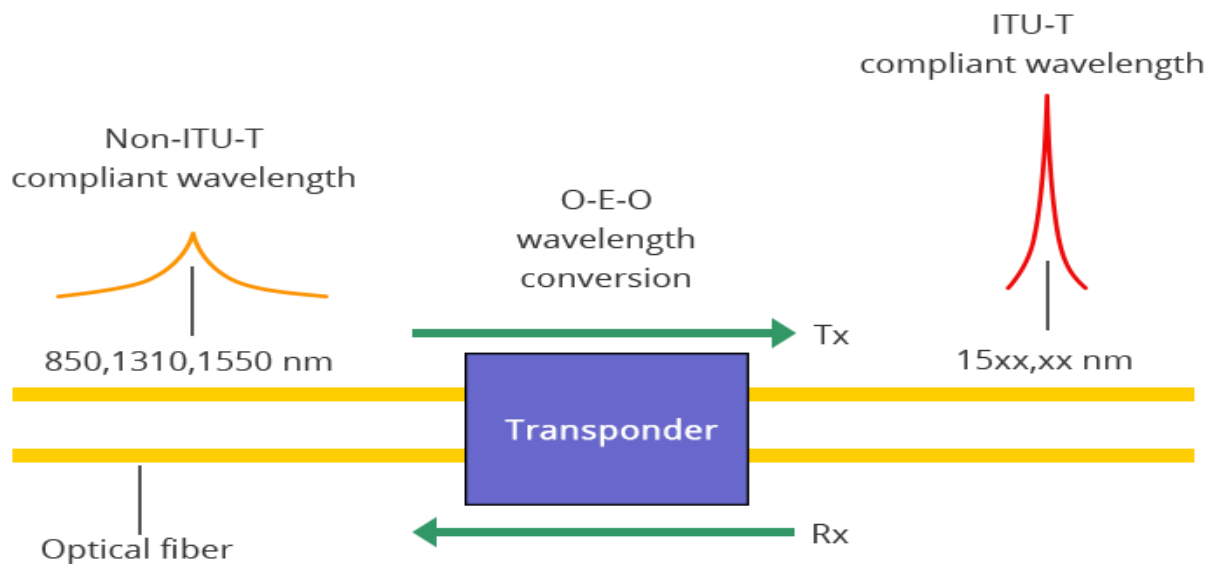
Several components work together to make up a DWDM optical system. Below is a high-level overview of each component and its function:

- **Transponder** – maps the data stream to an outgoing light wavelength that suits the DWDM application.
- **Muxponder** – maps several data streams to an outgoing light wavelength that suits the DWDM application.
- **Optical add/drop multiplexers (OADM)** – consolidates multiple wavelengths of light onto an optical fibre
- **Optical amplifiers** – boosts the signal through the fibre optic network.
- **Fiber optic cables** – data travels across the fibre optic network as wavelengths of light.
- **Optical Supervisory Channel** - a dedicated communication channel that runs alongside the main data-carrying wavelengths within the same optical fibre.
- **Receiving equipment** – the signal arrives at its destination, goes through another OADM to demultiplex it into different wavelengths, and moves through the transponder (or muxponder) and router to the receiving equipment (e.g. a television, computer, etc.)

5.3.1 Transponders (Wavelengths Converters)

Transponders convert optical signals from one incoming wavelength to another outgoing wavelength suitable for DWDM applications. Transponders are Optical-Electrical-Optical (O-E-O) wavelength converters. A transponder performs an O-E-O operation to convert wavelengths of light, thus some people call them "OEO" for short. Within the DWDM system, a transponder converts the client optical signal back to an electrical signal (O-E) and then performs either 2R (Reamplify, Reshape) or 3R (Reamplify, Reshape, and Retime) functions.

The figure below shows bi-directional transponder operation.



A WDM transponder is located between a client device and a DWDM system. From left to right, the transponder receives an optical bit stream operating at one particular wavelength (1310 nm). The transponder converts the operating wavelength of the incoming bitstream to an ITU-compliant wavelength. It transmits its output into a DWDM system. On the receiving side (right to left), the process is reversed. The transponder receives an ITU-compliant bitstream and converts the signals back to the wavelength used by the client device.

Transponders are generally used in WDM systems (10 to 400 Gbps), including not only DWDM systems but also CWDM systems.

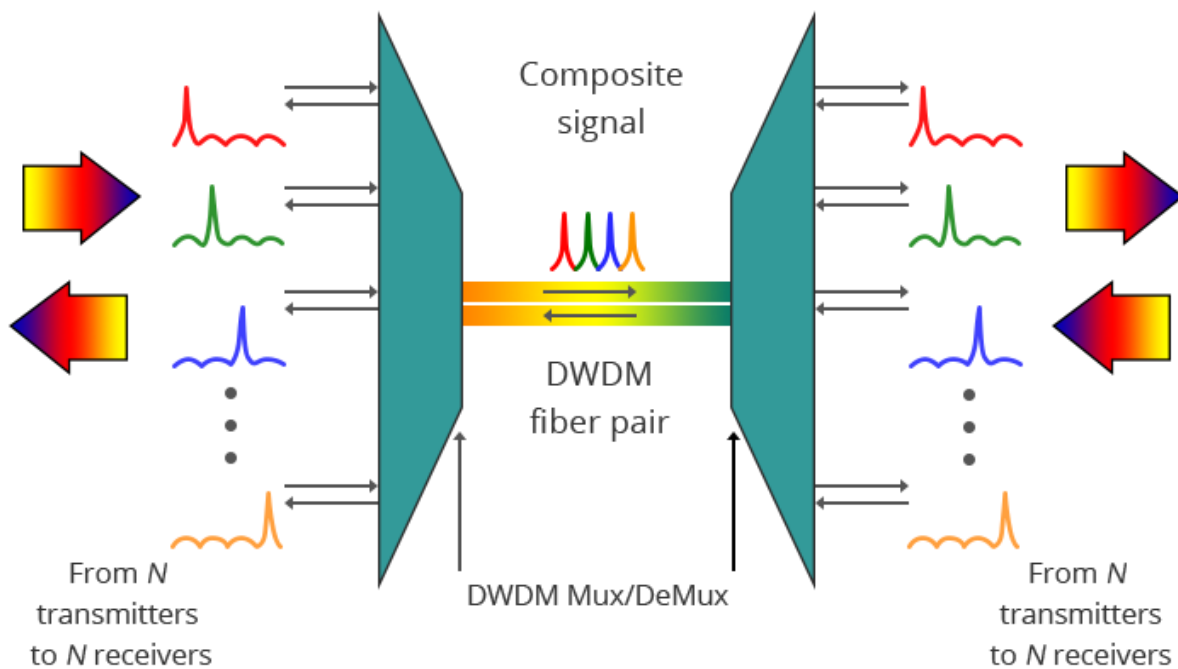
5.3.2 DWDM Multiplexers/Demultiplexers

DWDM Mux/Demux, integral to DWDM systems, serves as a critical passive optical component facilitating the simultaneous transmission of multiple data channels over a single fibre.

- The **Mux** combines diverse wavelengths onto the fibre, adhering to the ITU grid for standardized channel spacing.

- At the receiving end, the **Demux** separates these wavelengths, enabling the independent processing of individual data streams.

Its passive nature ensures low insertion loss and advanced configurations may include add-drop functionality for increased network flexibility. Typically, Mux and Demux (transmit and receive) components are contained in a single enclosure. Optical Mux/DeMux devices can be passive. Component signals are multiplexed and demultiplexed optically, not electronically, therefore no external power source is required. The figure below is a bidirectional DWDM network operation.



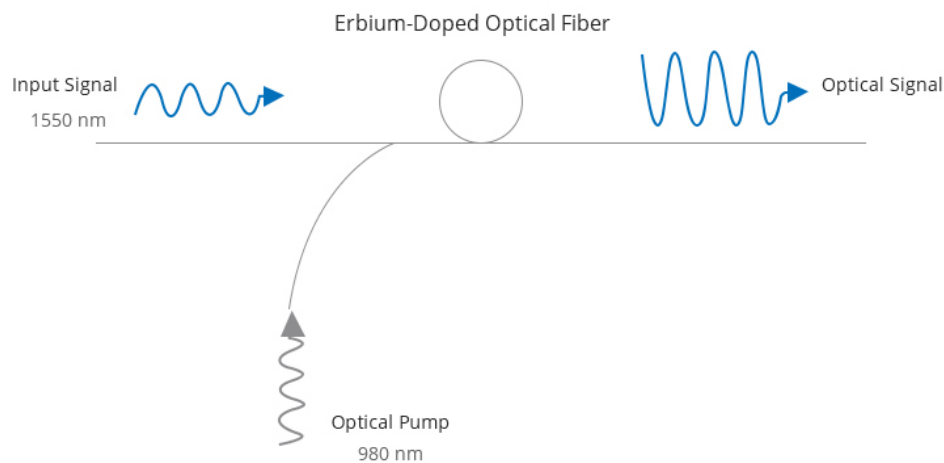
N light pulses of N different wavelengths carried by N different fibres are combined by a DWDM Mux. The N signals are multiplexed onto a pair of optical fibres. A DWDM Demux receives the composite signal and separates each of the N component signals and passes each to a fibre. The transmitted and received signal arrows represent client-side equipment. This requires the use of a pair of optical fibres; one for transmission, one for receive.

5.3.3 Amplifiers

Optical amplifiers boost the amplitude or add gain to optical signals passing on a fibre by directly stimulating the photons of the signal with extra energy. They are "in-fibre" devices. Optical amplifiers amplify optical signals across a broad range of wavelengths. This is very important for DWDM system applications. Erbium-Doped Fiber Amplifiers (EDFAs) are the most commonly used type of in-fibre optical amplifiers. EDFAs used in DWDM network are sometimes called DWDM EDFA, compared to

those used in CATV or SDH systems. To extend the transmission distance of your DWDM system you can choose from different types of optical amplifiers, including DWDM EDFA, CATV EDFA, SDH EDFA, EYDFA, and Raman Amplifier etc.

5.3.3.1 EDFA



The optical signal, such as a 1550 nm signal, enters an EDFA amplifier from the input.

- The 1550 nm signal is combined with a 980 nm pump laser with a WDM device—the signal and the pump laser pass through a length of fibre doped with Erbium ions. As discussed above, EDFA uses the erbium-doped fibre as an optical amplification medium.
- The 1550 nm signal is amplified through interaction with the doping Erbium ions. This action amplifies a weak optical signal to a higher power, effecting a boost in signal strength.
- EDFA amplifier working principle involves using a pump laser to excite erbium ions within the fibre. When the incoming optical signal stimulates these excited ions, they release additional photons, thus amplifying the signal.

In summary, an EDFA works by using stimulated emission in an erbium-doped fibre to amplify optical signals. The pump laser excites erbium ions in the fibre, and when incoming signals stimulate these ions, additional photons are emitted, amplifying the original signals. This process is crucial in long-distance optical communication systems to compensate for signal attenuation.

5.3.3.2 Types of EDFA Optical Amplifiers

- **Optical Booster Amplifier (OBA)**
 - Fiber optic signal booster is often placed at the transmitter side to amplify the optical signal before it emits to the fibre link. Typically, it is installed closely after the Mux Demux equipment to compensate for the power loss of DWDM wavelengths or single

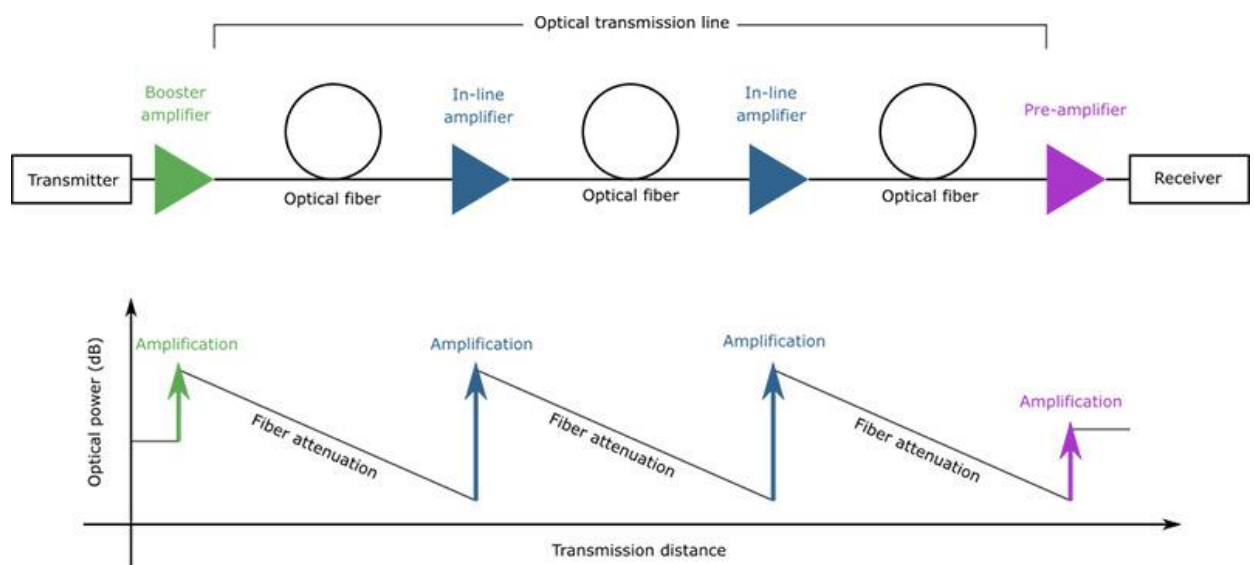
wavelengths. Booster EDFA is designed with low gain and high output power to be combined with WDM multiplexing and demultiplexing technologies.

- **In-line Amplifier (OLA)**

- As the name indicates, optical line amplifiers are put in the transmission line to compensate for the link attenuation caused by signals transmitted through long-haul fibre links. Whether installing an inline amplifier is necessary or how many to deploy depends on the fibre link distance. Generally, an optical line amplifier will be placed every 80km in a long-haul fibre link. Adding LAs to the optical line is a solution to extend the transmission distance. The LA amplifiers feature moderate or high gains and share the same high output power as the BA.

- **Preamplifier (OPA)**

- The optical preamp is located close to each of the receiver sides to guarantee sufficient optical power can be detected by the receiver. The OPA is necessary for amplifying the optical signals to a required level after they go through a long fibre journey. Market optical preamplifier often provides high gains and lower output power.

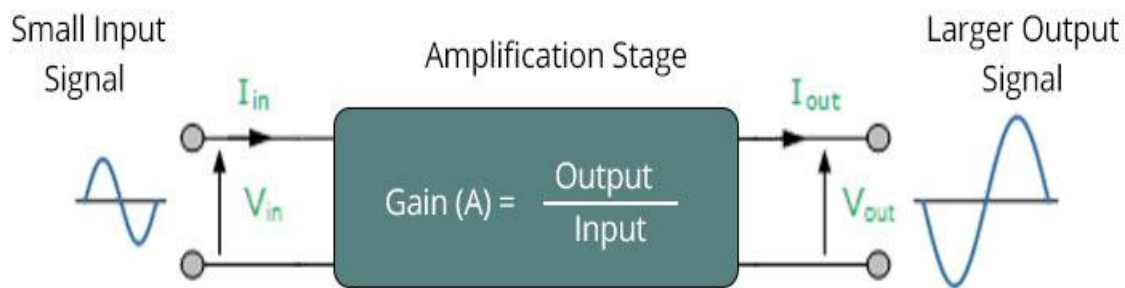


As the above diagram shows, each of the transmitting sides and the receiving sides are correspondingly installed with a booster amplifier and a pre-amplifier to perform signal amplification after or before the WDM node. Besides, one or several line amplifiers are placed along the fibre transmission line to compensate for the power loss brought by fibre attenuation. OBA and OPA are used in pairs in converged DWDM networks, however, the OLA is selective for adoption in real applications, please check the transmission distance.

5.3.3.3 EDFA Gain and Noise figures

The Gain of EDFA is the ratio of the optical output signal to the optical input signal. It is to measure the amplification degree the EDFA performs to the input signals. Depending on a different measured parameter of the optical power, the EDFA Gain can be Power Gain (**Ap**), Current Gain (**Ai**), or Voltage

Gain (**Av**). However, what we mentioned about optical gain often refers to Power Gain. The input signal and output signal before and after going through the amplifier are shown in the picture below



Optical Voltage Gain = Output Voltage / Input Voltage

Optical Current Gain = Output Current / Input Current

Since Power = Voltage * Current,

then:

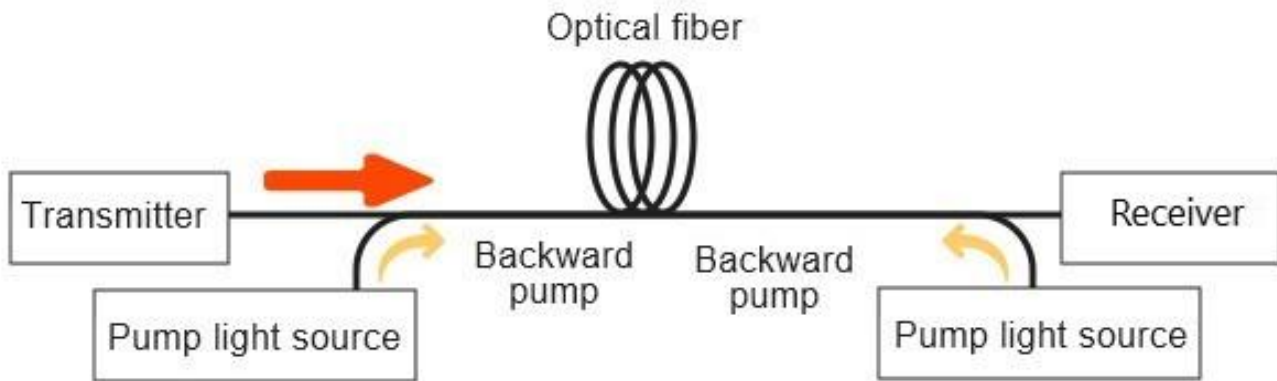
Optical Power Gain = Optical Voltage Gain * Optical Current Gain
 = (Output Voltage / Input Voltage) *(Output Current / Input Current)

EDFA Noise Figure, is the process of amplification generates Amplified Spontaneous Emission (ASE) noise, which will add to the signal and lead to a decrease in the signal-to-noise ratio (SNR) of the output. The ratio of input SNR to the output SNR is specified as Noise Figure (NF):

Noise Figure (NF) = Input SNR / Output SNR

5.3.3.4 Raman Amplifier

DWDM Raman amplifiers leverage **stimulated Raman scattering (SRS)** to amplify optical signals across multiple wavelengths in dense wavelength-division multiplexing (DWDM) systems. Unlike Erbium-Doped Fiber Amplifiers (EDFAs), Raman amplifiers use the transmission fibre itself as the gain medium, enabling distributed amplification and improving noise performance



- **Principles:**

- Pump lasers inject high-power light (typically 13xx–15xx nm) into the fiber, transferring energy to longer-wavelength signals via SRS. This creates a **broadband gain spectrum** (up to 100 nm bandwidth)

- **Directionality:**

- **Backward pumping:** Most common, where pumps counter-propagate signals to reduce noise and nonlinear effects.
- **Forward pumping:** Rarely used due to higher noise.

5.3.3.5 Key Advantages in DWDM Networks

1. **Distributed Gain:**

- Amplification occurs along the fibre span, reducing signal attenuation and improving optical signal-to-noise ratio (OSNR) by 3–6 dB compared to EDFA-only systems.

2. **Noise Reduction:**

- Lower noise figure (e.g., -5 dB in some configurations) due to distributed amplification, enabling longer spans (up to 150 km).

3. **Flexibility:**

- Gain spectrum can be tailored by adjusting pump wavelengths and power, supporting C-band (1530–1565 nm) and L-band (1570–1610 nm).

4. **Hybrid Systems:**

- Combining **Raman + EDFA** provides higher OSNR and reach than EDFA alone. For example, a 16-channel DWDM system with EDFA+Raman achieved **21.46 Q-factor** vs. 20.20 for Raman+EDFA at 80 km

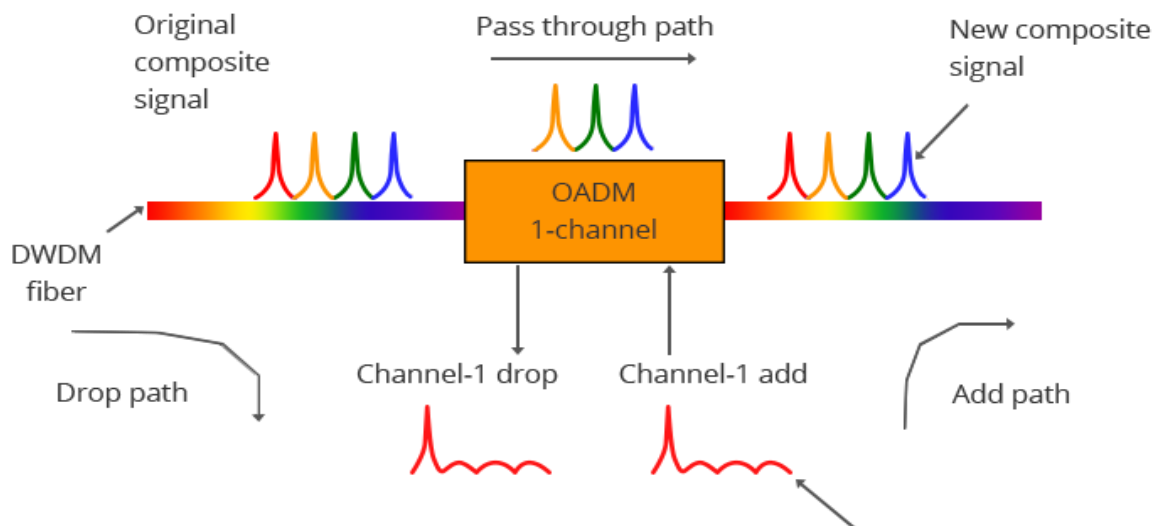
5.3.3.6 Comparison of EDFA vs RAMAN Amplifiers

Feature	Raman Amplifier	EDFA
Gain Medium	Transmission fibre	Erbium-doped fibre
Noise Figure	Lower (e.g., -5 dB)	Higher (e.g., 4–6 dB)
Gain Bandwidth	Broadband (~100 nm)	Limited to C/L-bands

Power Efficiency	Requires high pump power (e.g., 250 mW)	More efficient (lower pump power)
Span Extension	Up to 150 km with hybrid systems	Typically, 80–100 km

5.3.4 Optical Add/Drop Multiplexers

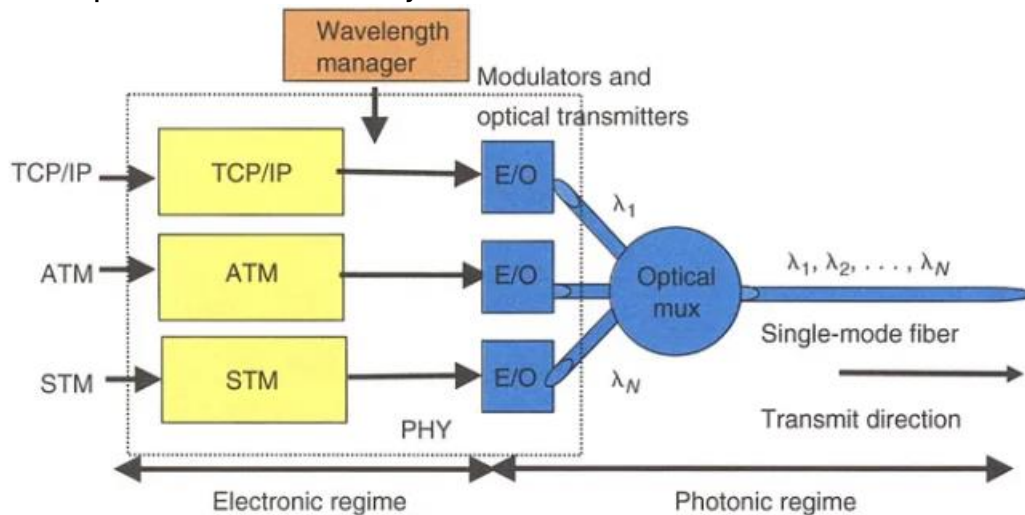
Optical add/drop multiplexers (i.e. **OADM**s) have a different function of "Add/Drop", compared with Mux/Demux filters.



Here above figure that shows the operation of a 1-channel DWDM OADM. This DWDM device is designed to only add or drop optical signals with a particular wavelength. From left to right, an incoming composite signal is broken into two components, drop and pass-through. The OADM drops only the red optical signal stream. The dropped signal stream is passed to the receiver of a client device. The remaining optical signals that pass through the OADM are multiplexed with a new add signal stream. The OADM adds a new red optical signal stream, which operates at the same wavelength as the dropped signal. The new optical signal stream is combined with the pass-through signals to form a new composite signal.

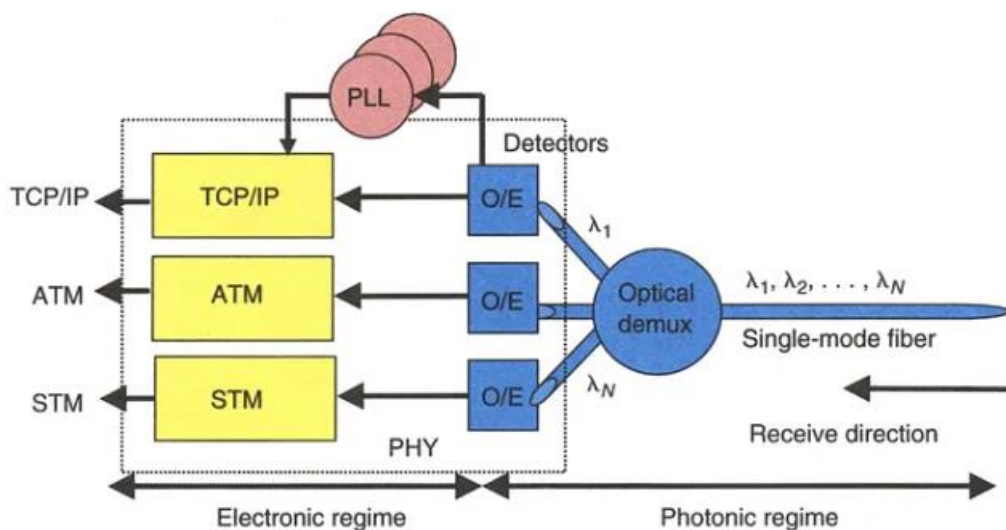
Transmit Direction

A DWDM hub accepts various electrical payloads, such as communications transport protocols/Internet Protocol (TCP/IP), asynchronous transfer mode (ATM), STM, and high-speed Ethernet (1 Gb/s, 10 Gb/s). Each traffic type (channel) is sent to its corresponding physical interface, where a wavelength is assigned and is modulated at the electrical-to-optical converter. The optically modulated signals from each source are then optically multiplexed and launched into the fiber.



Receive Direction

When a hub receives a WDM signal, it optically demultiplexes it to its component wavelengths (channels) and converts each optically modulated signal to a digital electrical signal. Each digital signal then is routed to its corresponding electrical interface: TCP/IP, ATM, STM, and so on. However, that each channel requires its own clock recovery circuitry because all channels may be at different bit rates.



Reconfigurable optical add-drop multiplexers (ROADMs) are now extensively employed, allowing the routing and rerouting of any wavelength in any direction. **ROADM** utilizes advanced next-generation wavelength-selective switch (WSS) technology to enable full CDC-F system configuration, featuring colourless, directionless, and contention less capabilities. With built-in pre-amplifier and booster amplifier, it simplifies installation and scalability while enhancing network flexibility and performance.

5.3.5 Optical Supervisory Channel (OSC)

A Supervisory Channel, specifically called the Optical Supervisory Channel (OSC) in DWDM (Dense Wavelength Division Multiplexing) systems, is a dedicated communication channel that runs alongside the main data-carrying wavelengths within the same optical fiber.

Its primary purpose is to carry network management, monitoring, and maintenance information—not user payload traffic

5.3.5.1 How does an Optical Supervisory Channel (OSC) work (methods)

A. Wavelength Allocation:

The OSC uses a specific wavelength that is *outside* the main data transmission band (typically at 1510 nm, 1620 nm, 1310 nm, or a proprietary wavelength), so it does not interfere with user data channels

B. Span-by-Span Communication:

The OSC is terminated and regenerated at every network node (such as amplifiers, ROADMs, or repeaters) along the DWDM route. This allows each node to insert its own local management information and forward the updated OSC signal to the next node

The optical supervisory channel are assigned a wavelength that can be located outside the usable gain bandwidth of the Amplifier ("out-of-band OSC") or alternatively, within the usable gain bandwidth ("in-band OSC"). There are design trade-offs associated with each of these possible choices.

C. Signal Format:

The **Optical Supervisory Channel (OSC)** in DWDM networks typically uses the **OC-3/STM-1 protocol structure** for its signal format, as recommended by ITU-T G.692 and implemented by major vendors. An optical service channel (OSC) is a bidirectional channel connecting two adjacent nodes in a DWDM network,

The frame structure of the OSC signal is like the OTN frame structure , Each frame has 4 x 4080 bytes.

	1...6	7	8...10	11...15	16...3824	3825...4080
1	FAS	RES	SM	RES	Payload (1588, DCN, FE_DCN, Panel FE)	FE C
2	RES					
3						
4						

FAS: Frame Alignment Signal
 SM: Section Monitoring
 RES: Reserved

OSC byte assignment

- Protocol**

Structure:

The OSC is most commonly formatted as an **OC-3 (Optical Carrier level 3) or STM-1 (Synchronous Transport Module level-1)** signal. This means it uses the same framing and overhead structure as SONET (for OC-3) or SDH (for STM-1) systems. (SONET/SDH are type of Optical Network which was used before DWDM evolved)

- Overhead Bytes:**

The **SDCC (Section Data Communication Channel) or RS-DCC (Regenerator Section Data Communication Channel)** overhead bytes in the OC-3/STM-1 frame are used for network management communication between nodes

- Payload Use:**

The payload portion of the OSC can be used to carry additional management data, such as Fast Ethernet frames, orderwire (voice/data for maintenance), alarms, configuration information, and remote software updates,

Common Vendor Implementations for OSC Payloads:

- Fast Ethernet or Gigabit Ethernet frames**

- This allows the OSC to transport IP-based management protocols (like SNMP, Telnet, SSH, or even web-based GUIs) for remote configuration and monitoring.
- Ethernet in the OSC payload is particularly useful for integrating with open-source controllers (e.g., OpenConfig, TransportPCE) and for supporting multi-vendor environments

- Orderwire and Voice/Data Channels**

- Vendors often implement orderwire (voice or low-speed data channels) in the OSC payload, enabling field technicians to communicate between sites for maintenance or troubleshooting

- Alarms, Configuration, and Remote Software Updates**

- The OSC payload is used to transport:
 - Alarm and fault information
 - Configuration commands

- Remote software/firmware updates
- Performance monitoring data (e.g., chromatic dispersion, PMD, OSNR)
- **Control Plane and Automation Support**
 - In advanced networks, the OSC payload may carry protocols for **ASON/GMPLS control plane signalling**, supporting dynamic provisioning, automated restoration, and mesh network management
 - Some vendors, especially in open networking environments (like Open ROADM MSA and IOWN GF), use the OSC to extend **OpenConfig** or similar APIs for direct device management and automation
- **Capacity:**

The OSC channel typically provides a capacity of at least **2 Mbps** (as per TEC and ITU-T G.692 requirements), sufficient for management and monitoring tasks
- **Vendor Specifics:**

While OC-3/STM-1 is a common standard, the actual signal structure and additional features may be **vendor-specific**. Some vendors may use proprietary enhancements or allow the OSC to carry Ethernet or other management protocols within the payload
- **Span-by-Span Termination:**

The OSC is terminated at every amplifier or network node, where it is converted from optical to electrical, processed, and then regenerated for the next span. This enables each node to insert local management information and alarms

Feature	Typical Implementation
Protocol/Format	OC-3/STM-1 (SONET/SDH)
Overhead Use	SDCC/RS-DCC bytes for management
Payload Use	Ethernet, orderwire, alarms, config data
Capacity	2 Mbps
Standard Reference	ITU-T G.692
Vendor Enhancements	Possible (e.g., Ethernet in payload)

5.3.5.2 Key functions of OSC

A. Network Management:

The OSC transmits essential management and control information, enabling centralized or distributed network management systems (NMS) to monitor and control the DWDM network.

B. Alarm and Fault Reporting:

It carries alarms, status updates, and fault information from remote amplifier or terminal sites to the NMS, allowing for rapid fault detection and troubleshooting

C. Performance Monitoring:

The OSC can carry information about optical signal quality, such as chromatic dispersion (CD), polarization mode dispersion (PMD), and optical power levels. This enables real-time health checks of each fiber span

D. Orderwire and Communication:

It supports voice or data channels (orderwire) for technical staff to communicate between sites during maintenance or troubleshooting

E. Remote Software Management:

The OSC can be used for remote software upgrades and configuration changes at amplifier or terminal nodes

5.3.5.3 Why is OSC important?

A. No Extra Fiber Needed:

Since the OSC uses a dedicated wavelength on the same fiber as the data channels, it eliminates the need for separate management fibers, saving infrastructure costs and simplifying deployment

B. Span-by-Span Visibility:

By being regenerated at every node, the OSC provides detailed, localized monitoring and control, which is essential for large-scale, long-haul, or complex DWDM networks

C. Supports Automation:

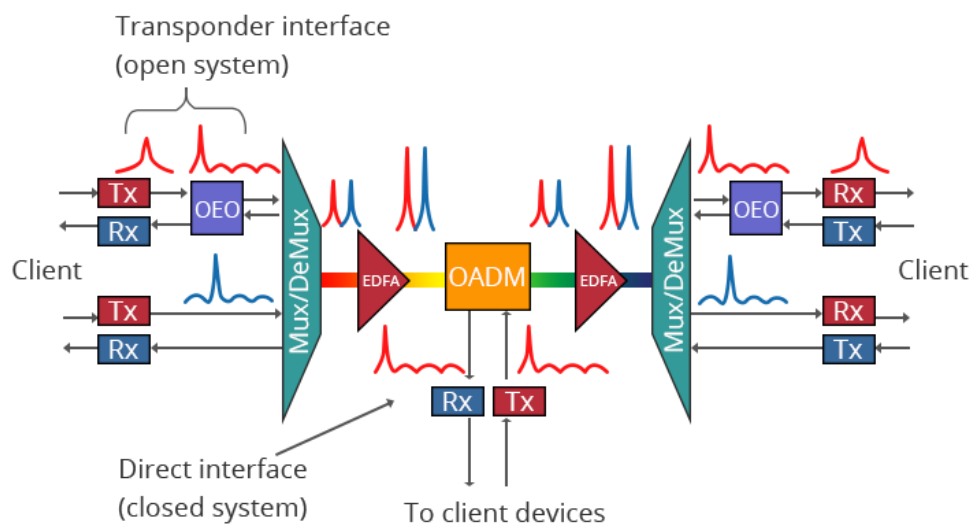
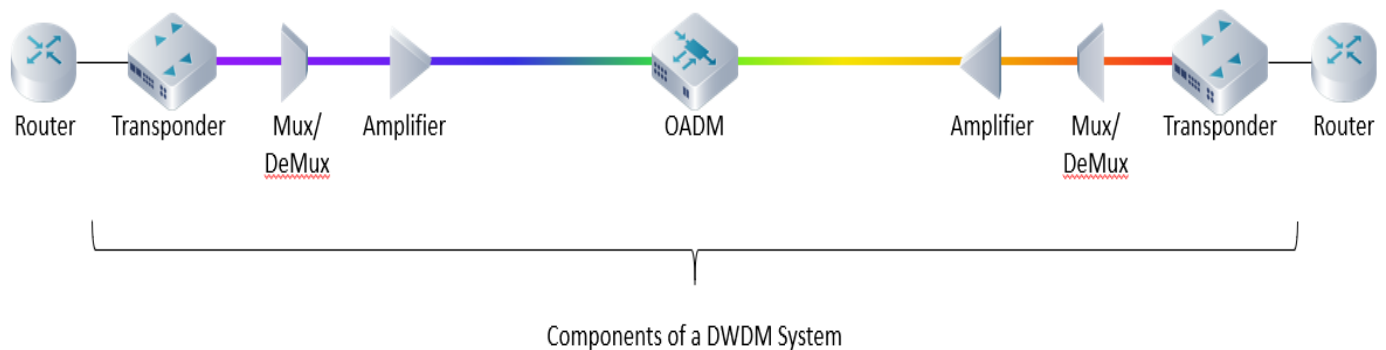
The OSC enables advanced features like automatic protection switching, dynamic provisioning, and real-time network optimization.

Feature	Description
Wavelength	1510 nm, 1620 nm, 1310 nm, or proprietary
Traffic Type	Management, monitoring, alarms, orderwire (not user data)
Span-by-Span Regeneration	Yes, at every node
Signal Format	OC-3/STM-1, Fast Ethernet, Gigabit Ethernet (vendor-specific)
Functions	Alarms, monitoring, remote management, performance data, orderwire
Integration	Amplifier/ROADM/terminal sites

The Supervisory Channel (OSC) is the management and monitoring “lifeline” of a DWDM network, ensuring operators have complete visibility and control over the health and operation of the optical transmission system—without impacting user data traffic.

While the OSC’s primary function is always network management and monitoring, many vendors enhance the payload by supporting Ethernet frames, orderwire, alarms, remote configuration, and even open APIs for automation and multi-vendor interoperability. The exact implementation depends on the vendor and network requirements, but Ethernet in the OSC payload is increasingly common in modern DWDM systems

5.4 How Does DWDM System Components Work Together with DWDM Technology?



- Data enters via the router and goes to the transponder.

- The transponder maps the signal to a DWDM wavelength and sends it to the multiplexer (Mux).
- Optical amplifiers boost the signal for longer distances after leaving the Mux.
- Optical add/drop multiplexers (OADM) add or remove bitstreams by wavelength, and additional amplifiers can further extend the distance.
- The signal is de-multiplexed (DeMux) into DWDM wavelengths, converted by the transponder, and routed to its destination.

5.5 DWDM Topologies

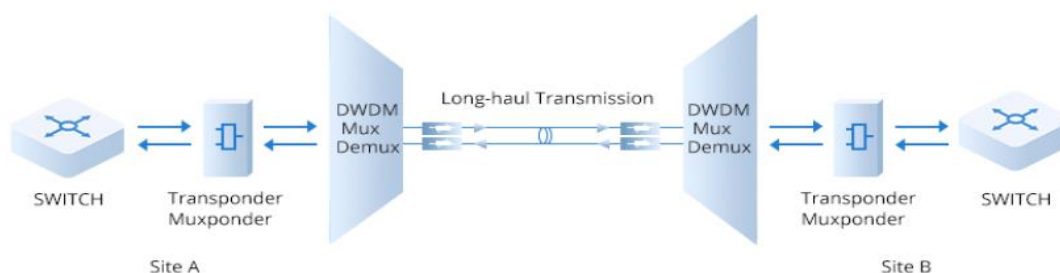
Dense wavelength division multiplexing (DWDM) networks are classified into four major topological configurations:

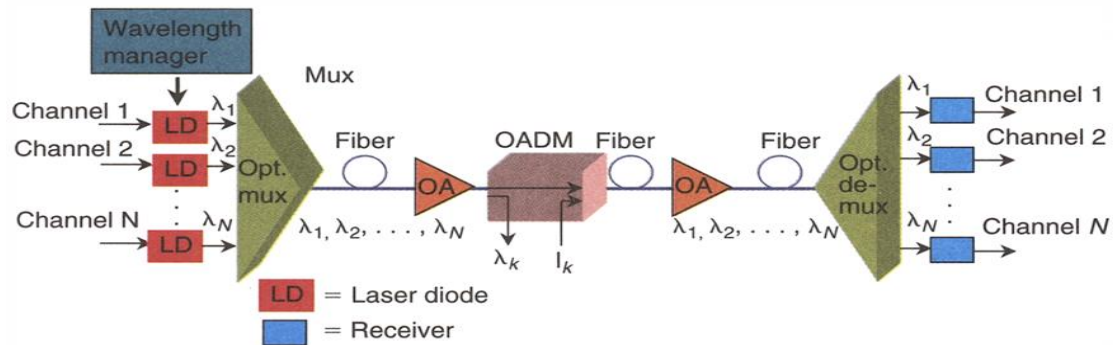
- DWDM point-to-point with or without add-drop multiplexing network
- Fully connected mesh network, star network, and DWDM ring network with OADM nodes and a hub. Each topology has its own requirements and based on the application, different optical components may be involved in the respective designs.
- In addition, there are hybrid network topologies that may consist of stars and/or rings that are interconnected with point-to-point links.

5.5.1 Point-to-point Topology

Point-to-Point (P2P) Topology in Dense Wavelength Division Multiplexing (DWDM) networks establishes direct connections between two endpoints, facilitating dedicated communication paths without intermediary nodes. This configuration ensures efficient data transmission, minimizing latency and enhancing security. P2P topology offers simplicity in network management, scalability for easy expansion, and cost-effectiveness due to its streamlined infrastructure.

The point-to-point topology in DWDM networks is primarily deployed in long-haul transport scenarios demanding exceptional aggregate bandwidth, superior signal integrity, high reliability, and rapid path restoration capabilities. Typically spanning several hundred kilometres, this topology features fewer than 10 amplifiers between transmitter and receiver endpoints.





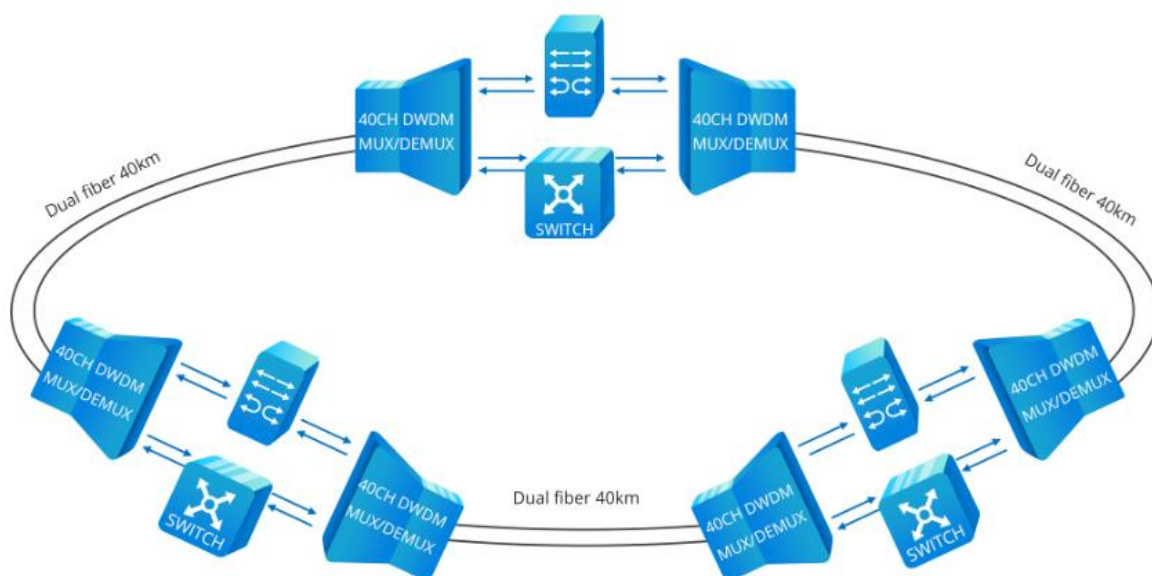
An end-to-end simplistic view of a DWDM point-to-point system that includes lasers, an optical multiplexer and demultiplexer, fibres, optical amplifiers (OA), and an optical add-drop multiplexer is shown in Figures above.

5.5.2 Ring Topology

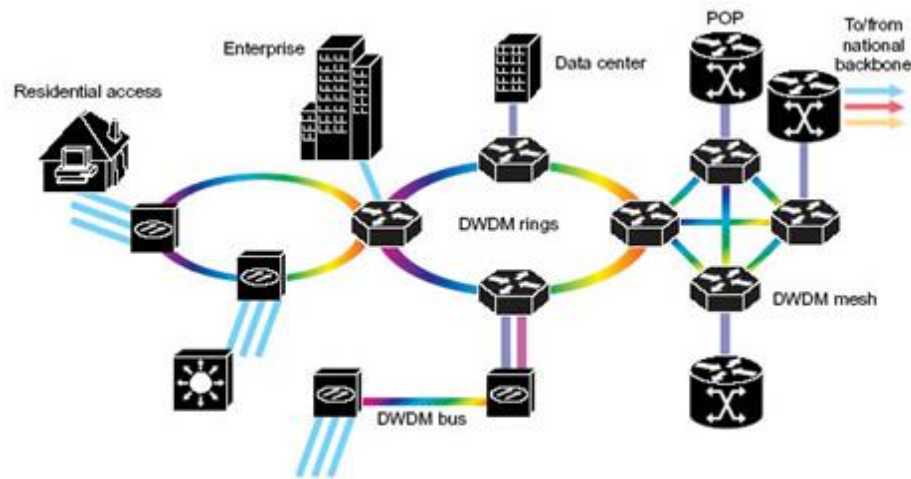
A DWDM ring network includes a fibre in a ring configuration that fully interconnects nodes. Two fibre rings are even presented in some systems for network protection. This ring DWDM topology is commonly adopted in a local or a metropolitan area which can span a few tens of kilometres. Many wavelength channels and nodes may be involved in the DWDM ring system

The main advantage of a Ring Topology is that it provides redundancy and fault tolerance. If one node or cable fails, the data can still travel around the ring in the opposite direction. A Ring Topology includes two fibre paths for network protection

In case of a fibre cut or a node failure, traffic can be switched to the backup path in a short time. A Ring Topology is suitable for local or metropolitan area applications that require high connectivity, high scalability, high flexibility, and high availability.



5.5.3 Mesh Topology



Mesh Topology in Dense Wavelength Division Multiplexing (DWDM) networks represents a highly interconnected configuration where each network node is directly connected to multiple other nodes, forming a complex web of interconnections. In DWDM, these connections are established using optical fibers to transmit data signals encoded as light pulses, with each node equipped with DWDM transceivers capable of transmitting and receiving data over multiple wavelengths.

Mesh topology offers redundancy and fault tolerance, as multiple paths exist between any pair of nodes, enabling data to be rerouted in case of link failures.

This topology enhances reliability and resilience, particularly in large-scale networks where maintaining continuous connectivity is critical.

However, mesh topology requires significant infrastructure investment and complex management due to the multitude of connections, making it more suitable for core networks or critical applications where high availability and fault tolerance are paramount.

5.6 DWDM Network Security / Protections

In fibre optic communications, wavelength division multiplexing (WDM) technology transports multiple optical carrier signals over a single optical fibre, enabling high traffic capacity.

Protection (multiple diverse paths for the fibre) is essential for many organizations to ensure the availability of the link and the data carried. A network outage or failure between sites brings tremendous business upheavals. In some industries, such as financial services, failures can also mean fines or legal action.

Data is the most important asset organizations have and must always be available throughout the organization. They cannot afford to risk network outages due to equipment or fibre link failures. Safeguarding data and network operations is of the highest importance and is done by building protection as part of their business continuity procedures.

In typical protected WDM networking solutions two or more sites are connected by diverse paths, always providing extremely high availability of the network. However, fibre cuts can occur for many reasons, whether malicious or benign (such as building works, street maintenance, and carrier faults). Equipment failures are far less frequent, but they can occur.

3.4.1 Protection Vs Restoration

Protection is defined as the primary mechanism used to deal with a failure. It needs to be very fast (typically traffic should not be interrupted for more than 60 ms in the event of a failure of DWDM Payload networks). As a result, the protection routes usually need to be pre-planned so that traffic can be switched over from the normal routes on to the protection routes quickly.

Due to the speed requirements, this function is usually performed in a distributed way by the network elements without relying on a centralized management entity to coordinate the protection actions.

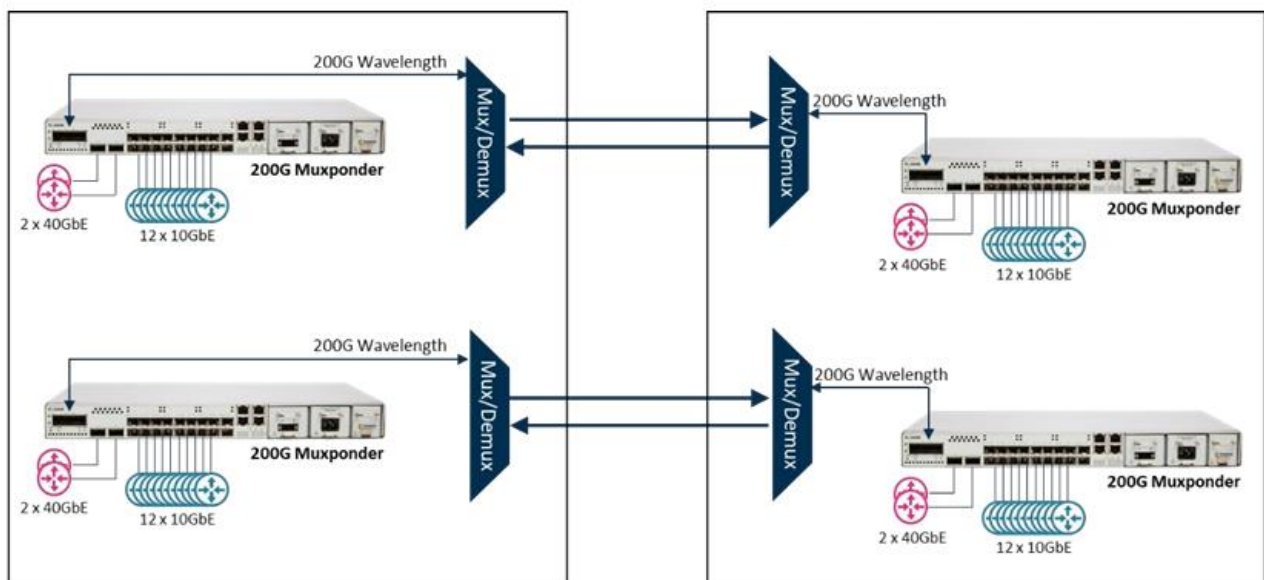
In contrast, **restoration** is not a primary mechanism used to deal with failure. After the protection function is complete, restoration is used to provide either efficient routes or additional resilience against further failures before the first failure is fixed. As a result, it can afford to be quite slow (seconds to minutes sometimes).

The restoration routes need not to be preplanned and can be computed on the fly by a centralized management system, without requiring a distributed control function. More sophisticated algorithms can be used to reduce the excess bandwidth required, and more complex mesh topologies can be supported.

Some of the protection methods that industry applies, each with its strengths and weaknesses as follows

3.4.2 Full Hardware Redundancy

This approach is also known as 1+1 full hardware redundancy. It involves two full sets of equipment at each site (see Figure 2). The equipment at each end consists of identical devices including muxponders/transponders, and protection is applied at the client switch level.

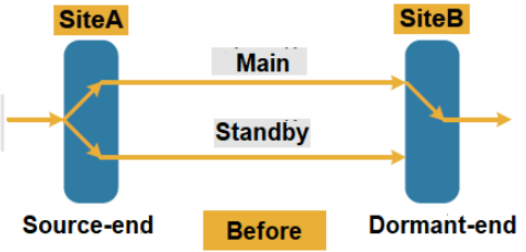
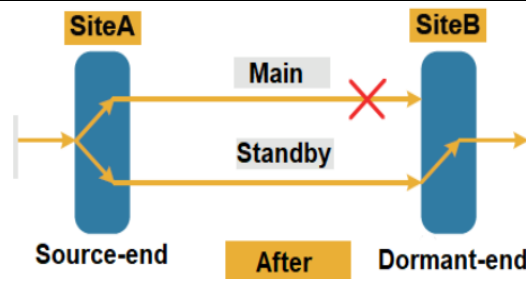


The same data traffic is delivered over two redundant links simultaneously at all times. While this approach can be more costly, it provides the best guarantee that the organization's traffic is fully protected. If there is a failure in any component in the network, or an interruption in the fibre link, the second complete link will continue to operate, giving the organization time to secure repairs on the other set or link.

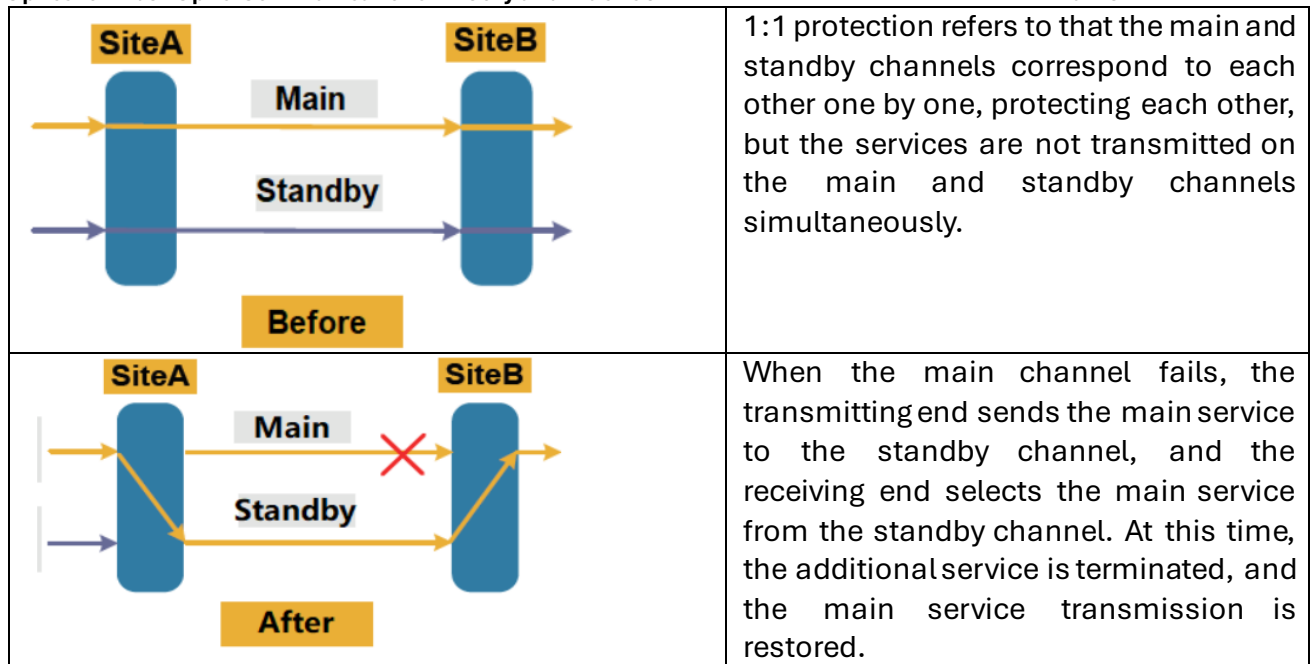
3.4.3 Protection per Wavelength

3.4.3.1 1+1 protection per wavelength

1+1 protection refers to that each main channel has an exclusive standby channel to provide protection for it.

	The transmitting end simultaneously sends data on the main and standby channels. Under normal circumstances, the receiving end selects the service on the main channel.
	When the main channel is abnormal, the receiving end selects the data of the standby channel according to the channel status and external commands, so that the service can be restored.

3.4.3.2 1:1 protection per wavelength



The transmitting end sends the main service on the main channel and sends the additional service (low-level service) on the standby channel. The receiving end receives the main service from the main channel and the additional service from the standby channel. (M:N protection: that is, M standby channels provide protection for N channels, generally $M \leq N$.)

3.4.4 Facility Protection - Fiber Path Protection

The fibre protection must a redundancy fibre for protection. If the working fibre cut, the traffic will be switching to the protection fibre automatically. This is all channels protection. But in this fibre protection the redundancy fibre length must be similar to working fibre , must consider and calculate the power budget is suit with working fibre path.

